SC01

You are being invited to participate in a research study titled “Improving Risk Literacy.” This study is being done by Dr. Vera Wilde. You were selected to participate in this study because you were on Prolific.

The purpose of this research study is to understand how different ways of presenting statistical information affect people’s reasoning about screening programs. If you agree to take part in this study, you will be asked to complete an online survey about interpreting test results in various screening contexts. This survey will take you approximately 30 minutes to complete.

You may not directly benefit from this research; however, we hope that your participation in the study may help improve how statistical information is communicated to the public, particularly in medical, security, and educational contexts.

We believe there are no known risks associated with this research study; however, as with any online activity the risk of a breach of confidentiality is always possible. To the best of our ability your answers in this study will remain confidential. We will minimize any risks by storing all data securely, using anonymous participant IDs, and disposing of identifying information after data collection is complete. All data will be stored in accordance with GDPR requirements.

Your participation in this study is completely voluntary and you can withdraw at any time. You are free to skip any question that you choose.

If you have questions about this project or if you have a research-related problem, you may contact the researcher: Dr. Vera Wilde at wilde.k.vera@gmail.com.

By clicking “I agree” below you are indicating that you are at least 18 years old, have read and understood this consent form and agree to participate in this research study.

- ○ No, I do not agree (do not participate in this study)
- ○ Yes, I agree

PR01

1. Prolific ID

Please enter your unique Prolific ID.

PHPcode

```
// Show shared instructions first
text('IN01');

// Randomize to Control or Experimenta
$condition = random(1, 2);

// Store which condition they got
if ($condition == 1) {
    put('IV01_01', 1); // Control
} else {
    put('IV01_01', 2); // Experimental
}

// Show the appropriate condition cont
if ($condition == 1) {
    question('CT01'); // Control: prob
} else {
    question('EX02'); // Experimental:
}
```

text('IN01')

Screening Programs and You

Screening programs are everywhere. They're used to keep us safer at airports, in workplaces, and in schools. They help monitor health throughout childhood, pregnancy, and old age. And they play a growing role in detecting fraud, spam, and misinformation online.

Across these different areas, screening programs share a similar structure: they check large numbers of people or items to find relatively rare problems. With increasing digitalization and artificial intelligence, these systems now affect millions of people every day.

In this session, you'll learn how screening programs work in practice and then answer some questions about interpreting their results.

We are interested in what and how people answer the questions. Please respond based on your own thinking, without any outside assistance.

q u e s t i o n (' C T 0 1 ')

CT01

Understanding Base Rate Bias in Mass Screening Programs

Base rate bias occurs when we ignore how common or rare a problem is (its "prevalence" or "base rate") when interpreting screening results. This leads to systematic errors in reasoning about mass screening programs in medicine, security, and other domains.

THE MATHEMATICS: BAYES' THEOREM

To correctly interpret a positive screening result, we need to use Bayes' theorem:
P(Problem | Flagged) = [P(Flagged | Problem) × P(Problem)] / P(Flagged)
Where:

P(Problem) = prevalence/base rate
P(Flagged | Problem) = sensitivity (true positive rate)
P(Flagged) = total probability of being flagged

EXAMPLE 1: Polygraph ("Lie Detector") Screening — Suspicious Mode

Consider screening scientists at national labs for spies using a polygraph set to maximize detection (high sensitivity, more false positives):

Prevalence: 0.1% are actual spies
Sensitivity: 80% (probability of being flagged if actually a spy)
False positive rate: 16% (probability of being flagged if not a spy)

Calculation:
P(Spy | Flagged) = (0.80 × 0.001) / [(0.80 × 0.001) + (0.16 × 0.999)]
P(Spy | Flagged) = 0.0008 / (0.0008 + 0.15984)
P(Spy | Flagged) = 0.0008 / 0.16064 = 0.5%

Result: An employee who is flagged has only a 0.5% probability of actually being a spy. For every real spy caught, approximately 200 innocent employees are falsely accused.

EXAMPLE 2: Polygraph ("Lie Detector") Screening — Friendly Mode

Now consider the same screening using a polygraph set to minimize false accusations (low sensitivity, fewer false positives):

Prevalence: 0.1% are actual spies
Sensitivity: 20% (probability of being flagged if actually a spy)
False positive rate: 0.39% (probability of being flagged if not a spy)

Calculation:
P(Spy | Flagged) = (0.20 × 0.001) / [(0.20 × 0.001) + (0.0039 × 0.999)]
P(Spy | Flagged) = 0.0002 / (0.0002 + 0.003896)
P(Spy | Flagged) = 0.0002 / 0.004096 = 4.9%

Result: An employee who is flagged has a 4.9% probability of actually being a spy — better than Suspicious Mode. But 80% of real spies are now missed (false negatives).

EXAMPLE 3: Mammography Screening

Consider breast cancer screening:

Prevalence: 0.5% of women have breast cancer
Sensitivity: 90% (probability of positive result if cancer present)
False positive rate: 10% (probability of positive result if no cancer)

Calculation:
P(Cancer | Positive) = (0.90 × 0.005) / [(0.90 × 0.005) + (0.10 × 0.995)]
P(Cancer | Positive) = 0.0045 / (0.0045 + 0.0995)
P(Cancer | Positive) = 0.0045 / 0.104 = 4.3%

Result: A woman who tests positive has only a 4.3% probability of actually having breast cancer.

EXAMPLE 4: PSA Screening (Prostate Cancer)

Consider prostate cancer screening:

Prevalence: 1% of men have prostate cancer
Sensitivity: 80% (probability of positive result if cancer present)
False positive rate: 15% (probability of positive result if no cancer)

Calculation:
P(Cancer | Positive) = (0.80 × 0.01) / [(0.80 × 0.01) + (0.15 × 0.99)]
P(Cancer | Positive) = 0.008 / (0.008 + 0.1485)
P(Cancer | Positive) = 0.008 / 0.1565 = 5.1%

Result: A man who tests positive has only a 5.1% probability of actually having prostate cancer.

EXAMPLE 5: Prenatal Screening (DiGeorge Syndrome)

Consider prenatal screening:

Prevalence: 0.04% of pregnancies have DiGeorge Syndrome
Sensitivity: 90% (probability of positive result if condition present)
False positive rate: 1% (probability of positive result if no condition)

Calculation:
P(Condition | Positive) = (0.90 × 0.0004) / [(0.90 × 0.0004) + (0.01 × 0.9996)]
P(Condition | Positive) = 0.00036 / (0.00036 + 0.009996)
P(Condition | Positive) = 0.00036 / 0.010356 = 3.5%

Result: A pregnancy that screens positive has only a 3.5% probability of actually having DiGeorge Syndrome.

KEY TAKEAWAY: When the base rate (prevalence) of a problem is low, even highly accurate screening programs produce a high proportion of false positives. The probability that being flagged indicates a true case depends critically on how rare the problem is. Always consider the base rate when interpreting screening results.

question('EX02')

EX02

Understanding Base Rate Bias in Screening Programs

You will now use an interactive tool called "Rarity Roulette" to explore how screening programs work when conditions are rare.

Please use the tool at: verawilde.github.io/rarity-roulette/

What to do:
Start with the security screening example (polygraphs, aka "lie detectors").
Then try the mammography, PSA testing, and prenatal testing examples.

For each, notice:
How many people have the condition (base rate)?
How many test positive overall?
The ratio of true positives to false positives?

Try adjusting the inputs (population, base rate, accuracy, and detection threshold) to see how changing parameters affects outcomes.
Spend approximately 15 minutes exploring.

Key things to notice:
When conditions are rare, even accurate tests produce many false positives.
There's always a trade-off between catching true cases and avoiding false alarms.
The frequency display ("8 out of 103") makes the math clearer than percentages.

When you're done exploring, click Next to continue.

TR01

Now you'll answer some questions about screening scenarios.

TP01

2. Mammography Screening

Out of 1,000 women who participate in routine mammography screening:
- 5 have breast cancer
- Of those 5 with breast cancer, 5 will test positive
- Of the 995 without breast cancer, 100 will also test positive

Out of the women who test positive, how many actually have breast cancer?

ANSWER FORMAT:
First line: Your answer as "_ out of _"
Then: Show your work (explain how you got your answer)

Example format:
30 out of 100
I calculated this by adding the true positives and false positives together, then finding how many were true positives.

TP02

3. Digital Communications Child Safety Screening

Out of 10,000 messages shared in a digital communications platform:
- 10 contain child sexual abuse material (CSAM)
- Of those 10 CSAM messages, 8 are detected by the AI screening system
- Of the 9,990 innocent messages, 999 are incorrectly flagged by the system

Out of the messages flagged by the screening system, how many actually contain CSAM?

ANSWER FORMAT:
First line: Your answer as "_ out of _"
Then: Show your work (explain how you got your answer)

Example format:
50 out of 200
I calculated this by adding the true detections and false flags together, then finding how many were true detections.

TP03

4. Prenatal Screening

Out of 7,000 pregnant women who receive prenatal screening:
- 10 are carrying a fetus with Down syndrome
- Of those 10, 9 will test positive
- Of the 6,990 not carrying a fetus with Down syndrome, 350 will also test positive

Out of the women who test positive, how many are actually carrying a fetus with Down syndrome?

ANSWER FORMAT:
First line: Your answer as "_ out of _"
Then: Show your work (explain how you got your answer)

Example format:
30 out of 100
I calculated this by adding the true positives and false positives together, then finding how many were true positives.

TP06

To confirm you are reading carefully, please type the word "banana" in the box below.

TP04

5. Academic Integrity Screening

Out of 10,000 student essays submitted in an online course:
- 100 essays contain AI-generated content that violates academic integrity policies
- Of those 100 essays with violations, 80 are flagged by the integrity detection system
- Of the 9,900 essays without violations, 990 are incorrectly flagged by the system

Out of the essays flagged by the detection system, how many actually contain violations?

ANSWER FORMAT:
First line: Your answer as "_ out of _"
Then: Show your work (explain how you got your answer)

Example format:
30 out of 100
I calculated this by adding the true violations and false flags together, then finding how many were true violations.

TP05

6. Social Media Content Moderation

Out of 100,000 posts on a social media platform:
- 200 posts contain hate speech that violates community guidelines
- Of those 200 posts with violations, 160 are flagged by the automated moderation system
- Of the 99,800 posts without violations, 4,990 are incorrectly flagged by the system

Out of the posts flagged by the moderation system, how many actually contain hate speech?

ANSWER FORMAT:
First line: Your answer as "_ out of _"
Then: Show your work (explain how you got your answer)

Example format:
30 out of 100
I calculated this by adding the true violations and false flags together, then finding how many were true violations.

BN01

7. Dice

Imagine we are throwing a five-sided die 50 times. On average, out of these 50 throws how many times would this five-sided die show an odd number (1, 3 or 5)?

BN02

8. Choir

Out of 1,000 people in a small town 500 are members of a choir. Out of these 500 members in the choir 100 are men. Out of the 500 inhabitants that are not in the choir 300 are men. What is the probability that a randomly drawn man is a member of the choir? Please enter your answer as a percentage (e.g., 25).

BN03

9. Loaded Die

Imagine we are throwing a loaded die (6 sides). The probability that the die shows a 6 is twice as high as the probability of each of the other numbers. On average, out of these 70 throws, how many times would the die show the number 6?

TN01

Finally, we'll ask a few questions about you.

SD01

10. What is your gender?

- ◯ female
- ◯ male

SD02

11. How old are you?

I am [ ] years old

SD07

12. Which country are you currently living in?

- ◯ Germany
- ◯ Austria
- ◯ Switzerland
- ◯ Other country: [ ]

SD11

13. What is your highest educational achievement?
Please select the highest level of qualification you have obtained.

- ◯ Less than high school
- ◯ High school diploma or equivalent
- ◯ Some college/university, no degree
- ◯ Associate’s/2-year degree
- ◯ Bachelor’s/undergraduate degree
- ◯ Master’s degree
- ◯ Doctoral or professional degree (PhD, MD, JD, etc.)
- ◯ Prefer not to say

SD19

To help us ensure data quality, please select option 4 below.

- [ ] Strongly disagree
- [ ] Disagree
- [ ] Neither agree nor disagree
- [ ] Agree
- [ ] Strongly agree

SD13

14. Are you currently employed?

- ◯ Yes, I am employed.
- ◯ No, I am unemployed.
- ◯ No, I am retired.
- ◯ No, I am a homemaker.
- ◯ No, none of the above.

SD14

15. What do you do professionally?

- ◯ Pupil/in school
- ◯ Training/apprenticeship
- ◯ University student
- ◯ Employee
- ◯ Civil servant
- ◯ Self-employed
- ◯ Unemployed/seeking employment
- ◯ Other: ____________

SD16

16. What is your monthly net income?

Net income is defined as your total income after tax and social security deductions.

[Please choose]

SD18

17. Would you like to comment this questionnaire, or would you like to add information for us to better understand your answers?

Do you think, this questionnaire needs improvement? Did you feel some question were unclear, or did you feel unpleasent answering specific questions? Please leave us notes.

SC02

18. You have answered all questions, thank you for your time! Can we use your data in anonymous form for scientific purposes?

- ◯ Yes, I have correctly answered all questions. My data can be used for the analyses.
- ◯ No, I wanted to “just look”, take part repeatedly, or do not want my data to be used for analyses.

SC03

Please be honest: Did you like participating this study? Please answer truely – your answer has no consequences for you!

- ◯ no
- ◯ rather not
- ◯ rather yes
- ◯ yes

SC04

Have you carried out all the tasks as described in the respective instructions? Please reply also honestly – this answer has no consequences for you!

- ◯ I have completed all the tasks as requested in the instructions.
- ◯ I <u>sometimes</u> clicked something because I was unmotivated or simply did not know.
- ◯ I have <u>often</u> clicked something, so I finish fast.

SC05

Could you complete the questionnaire in one without being distracted? Please reply honestly.

- ◯ Yes, I have given the study full attention and gone through it in one.
- ◯ I was distracted by my environment (people, noises, etc.) <u>once</u>.
- ◯ I was distracted by my environment (people, noises, etc.) <u>several times</u>.

Do you have a guess as to what the investigation was about? SC06

You can write down your thoughts briefly in the text field. If you have no idea, just leave the field blank.

Page 14

DB01

19. Thank You for Participating!

Thank you for completing this study on base rate reasoning in screening programs.

WHAT THIS STUDY WAS ABOUT:
This study examined whether frequency format presentations (like the interactive Rarity Roulette tool) help people understand base rate bias better than traditional probability format explanations.

Base rate bias occurs when we ignore how common or rare a condition is when interpreting test results. This is a widespread problem that affects decisions in medicine, security, education, and many other fields.

If you have any questions about this study, please contact: wilde.k.vera@gmail.com

Thank you again for your time and thoughtful responses!

Last Page

Thank you for completing this questionnaire!

We would like to thank you very much for helping us.

Your answers were submitted, you may close the browser window or tab now.